



Diya Gupta

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Bachelor of Technology

in Information Technology

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

Portfolio: <https://portfolio-indol-six-30.vercel.app/>

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EDUCATION

- B.Tech in Information Technology** Jul 2023 – May 2027
National Institute of Technology, Jalandhar CGPA: **9.1**
- Senior Secondary Education (Class XII), Science Stream** Apr 2023
Ch Balbir Singh Public School (CBSE) **96%**

EXPERIENCE

- Microsoft** June 2026 – July 2026
Software Engineering Intern Noida, India
 - Developed a **DNS telemetry framework** for Microsoft Defender on iOS, enabling DNS hostname visibility through the EDR pipeline and SOC Advanced Hunting.
 - Built a **Bloom filter-based telemetry solution**, achieving a 10^{-7} **false-positive rate** and reducing DNS payload size.
 - Benchmarked **CPU, memory, battery, and network usage** across raw DNS and Bloom filter approaches to quantify performance trade-offs.
 - Evaluated **Per-App VPN, Device-Wide VPN, and Control Filter** architectures across security visibility, telemetry quality, scalability, and performance.

PROJECTS

- PulmoAI: CT Scan & Pulmonary Nodule RAG Assistant** [Website](#)
Individual Project
 - Tech Stack:** Python, Flask, LangChain, Pinecone, HuggingFace (BGE), Groq API, HTML/CSS/JS
 - Built** a domain-specialized Retrieval-Augmented Generation (RAG) system to handle 100+ structured medical queries for CT scan and pulmonary nodule analysis.
 - Indexed** and chunked 500+ medical document chunks using Pinecone vector database and BGE embeddings for semantic search.
 - Integrated** Groq LLaMA model via API to generate guideline-aligned radiology responses with sub-second inference latency.
 - Implemented** a Flask-based chatbot interface supporting real-time interaction and concurrent request handling.
- ServEase: Full-Stack Service Marketplace** [Website](#)
Individual Project
 - Tech Stack:** Next.js 16, React 19, Node.js, Express.js, MongoDB Atlas, JWT, Tailwind CSS
 - Architected** and deployed a full-stack service marketplace connecting clients, vendors, and admins using Next.js App Router, React, and Node.js/Express.js.
 - Implemented** Role-Based Access Control (RBAC) using stateless JWT authentication and bcrypt password hashing across 3 user roles: Client, Vendor, and Admin.
 - Engineered** multi-parameter search with case-insensitive filtering by location, category, and price range using debounced client-side state management.
 - Developed** a Vendor Workspace with KPI metrics and admin moderation, backed by MongoDB Atlas, Mongoose, REST APIs, and CI/CD deployment using Vercel and Render.

TECHNICAL SKILLS

- Programming Languages:** C, C++, Python, JavaScript
- Tools, Libraries and Frameworks:** GitHub, Postman, Bootstrap, ReactJS, MySQL, ExpressJS, MongoDB Atlas
- Cloud and Platform:** Microsoft Azure
- Machine Learning Data:** Machine Learning, Data Preprocessing, Model Evaluation
- Course Fundamentals:** OOP, DBMS, CN, OS
- Soft Skills:** Strong team collaboration; experience leading workshops and coordinating cross-functional teams

POSITIONS OF RESPONSIBILITY

- Internship Representative** Jun 2025
Information Technology Department, NIT Jalandhar
- Co-Lead** Aug 2024
Cybernauts Community, NIT Jalandhar

ACHIEVEMENTS AND CODING PROFILES

- Academics:** Ranked **2nd** in academics in IT Dept., NIT Jalandhar 2025
- LeetCode:** Solved **600+** DSA problems, Contest Rating: 1520 [Profile](#) 2025
- GeeksForGeeks:** Solved **80+** DSA problems [Profile](#) 2025
- Hackmol 5.0:** Among Top **5** teams (Freshers Track) in national hackathon [Devfolio Link](#) 2024